



Speaker Series

Fall 2026 · Professional Development

About This Deck

This is the Professional Development committee's shortlist of talk ideas for the pledge class this fall.

About two-thirds of the class wants to go into software engineering and about a third into finance or quant, with a few people leaning toward health tech. We looked for topics that are specific, recent, and not something you'd get from a careers page, then figured out who could actually give each one.

Every idea comes with one question worth asking that speaker. At the end we rank our favorites and cover how we'll reach out.

Have a speaker lead or an idea we missed? Send it to the PD chairs.



Software Engineering

01



Finance & Quant

02



Health Tech

03



Top Picks

04



Next Steps

05

01

Software Engineering

- Systems & Scale
- Low-Level & Performance
- SWE + AI
- Weird & Fun

SWE: Systems & Scale

AEZ

What breaks when millions of people hit the same system at once

OUR PICK



01

The DNS Bug That Broke the Internet

In Oct 2025, one empty DNS record inside AWS took down Venmo, Snapchat, Fortnite and Canvas.

Book SRE or incident commander at a cloud provider

Ask What do you check first when everything is red?



02

One Database, 800 Million Users

OpenAI runs ChatGPT on one Postgres primary and about 50 read replicas, with no sharding.

Book Database or infra engineer at an AI company

Ask When do you finally give up and shard?



03

Moving Every Discord Message

Discord moved every message it stores to ScyllaDB and went from 177 database nodes to 72.

Book Storage engineer at a big consumer app

Ask How do you test a migration you only get to run once?

SWE: Low-Level & Performance

For the people who want to know what the computer is doing

AEZ



Rust Is in the Kernel Now

Linux 7.0 shipped in April with Rust officially out of "experimental." New drivers are being written in it.

Book Systems engineer on Android, Cloudflare or a kernel team

Ask *Where does Rust still get in your way?*



The Kernel Behind Every Chatbot

GPU kernels decide how fast and cheap AI gets, and most of the speedup comes from moving memory.

Book GPU performance engineer at NVIDIA or an AI lab

Ask *How do you know a kernel is maxed out?*



Why Your Code Is Slow

Reading flame graphs to find the one-line change that makes a service 10x faster.

Book Performance engineer at a trading firm or big tech

Ask *What's the biggest win you got from a tiny change?*

Key Takeaways

Start with Rust. It's the most concrete of the three, and anyone who has taken CS 61C or CS 162 will follow it. If an NVIDIA alum says yes to the GPU talk, take it. That's the hardest speaker here to get.

75%

of Google's new code
is written by AI

Sundar Pichai, April 2026. In late 2024 it was about a quarter, and Microsoft has said 20 to 30%. What we want to know is what the day-to-day looks like now.

OUR PICK



75% of Google's New Code Is Written by AI

Sundar Pichai's number from April 2026. In late 2024 it was about a quarter.

Book Staff engineer who led an AI coding rollout
Ask What do you look for in new grads now?



Testing an LLM Before Users Do

How teams catch a model update that breaks their product.

Book Applied AI engineer shipping LLM features
Ask What's an eval that caught something a human missed?



Why AI Keeps Getting Cheaper

Kimi K3 has 2.8T parameters but only runs 104B of them per token. Tricks like that keep pushing prices down.

Book Inference engineer at Fireworks or Baseten
Ask What's the real bottleneck now: chips, memory or power?

Weird & Fun SWE

No career payoff, just great stories

AEZ

Rebuilding Mario 64 From Scratch



The talk

Fans turned the original N64 game back into readable C code. Fan-made PC ports now run it natively, some at 60fps.

Who to book

Reverse engineer or security researcher

Ask them

How do you read code never meant to be read?

Why Fighting Games Rewind Time

The talk

Rollback netcode guesses your opponent's input, then rewinds a few frames when it guesses wrong. Street Fighter 6 runs on it.

Who to book

Netcode engineer at a game studio

Ask them

How many frames can you rewind unnoticed?

02

Finance & Quant

- Trading & Market Structure
- SWE × Finance

Trading & Market Structure

How trading firms make money, and the tech behind it

AEZ

OUR PICK



01

Microseconds Are Money

Light moves about 50% faster through air than through glass fiber, so trading firms built microwave towers from Chicago to New Jersey.

Book Low-latency engineer at Jump, HRT, Optiver or Citadel Securities

Ask Where does the next microsecond come from?



How Market Makers Actually Make Money

02

You can be right about the price and still lose money. Here's why.

Book Options trader at Optiver, IMC or SIG

Ask Ever been right and still lost money?



Betting on Everything

03

Kalshi and Polymarket traded \$53B in July, up from under \$5B last September. Most of it is sports.

Book Trader on an event-contracts desk

Ask Where does retail lose the most edge?

SWE × Finance

SWE jobs that touch money

AEZ



PAYMENTS

01

Charging a Card Exactly Once

Retries and idempotency keys, and why a double charge is fintech's worst bug.

Book Payments engineer at Stripe, Ramp or Brex

Ask *What's the worst money bug you've seen reach production?*



STABLECOINS

02

Stablecoins Got a Rulebook

The GENIUS Act became law in July 2025. This year banks and fintechs started building on it.

Book Digital-assets lead at a bank or fintech

Ask *Would you run payroll on stablecoins today?*

Key Takeaways

Payments wins this one. Every SWE will write a retry loop someday, and this talk shows what happens when it's wrong. The stablecoin talk is better for the finance people who want to understand what banks are building right now.

03

Health Tech

● — AI in the Exam Room

● — Regulated Software

● — Health Data

Health Tech

Software where a bug can hurt a patient

AEZ

OUR PICK



The AI That Writes Your Doctor's Notes

Epic launched AI Charting in February 2026. Abridge, already in 300+ health systems, is valued at \$5.3B.

Book Engineer at Abridge or a similar startup
Ask *What happens when it mishears a dosage?*



Software the FDA Signs Off On

About 1,500 AI devices are FDA-cleared, and most of them read medical images.

Book ML or regulatory engineer in medtech
Ask *How do you update a model the FDA already cleared?*



Your Medical Records Finally Have an API

Over 1,000 Epic hospitals are on TEFCA, the national records network, and FHIR APIs let apps pull a patient's records.

Book Interoperability engineer at Epic or a health-data startup
Ask *Why did this take until 2026?*

04

Top Picks

-  Ranking
-  Talk Spotlights

Our Top 5 for a Mixed Room

Ranked by how much of the whole class would care

AEZ

01 Microseconds Are Money

The one talk both the SWEs and the quants will care about.

02 75% of Google's New Code Is Written by AI

Everyone wants to know what this means for their first job.

03 Betting on Everything

10x growth in a year, with pricing and exchange tech in one talk.

04 The DNS Bug That Broke the Internet

Best story on the list. Most of us had an app go down that day.

05 The AI That Writes Your Doctor's Notes

Gets the health-tech people in, and Epic vs. Abridge is a real fight.

Talk Spotlight: Microseconds Are Money

AEZ

Firms built microwave towers from Chicago to New Jersey because air beats glass fiber

strategy.log

```
.000000533 md ask cleared  
.000000871 sig imbalance +0.81  
.000001152 ord BUY 5 @ 101.26  
.000019304 ack filled
```

tick to trade

to exchange

740 ns 18 μ s

Every step of a trade, timed in nanoseconds



What they'd cover

- Where the time goes: network, kernel, code, exchange
- FPGAs, kernel bypass, and why C++ still runs the floor
- How a firm decides a microsecond is worth building for



Who to book

- Low-latency engineer at Jump, HRT or Optiver
- Berkeley EECS alumni at these firms are the easiest ask

Ask Them

Where does the next microsecond come from?

Then: What did you build in your first month there?

Talk Spotlight: Google's 75% AI Code

AEZ

Most of Google's new code is now AI-generated. So what do engineers do all day?



What they'd cover

- A normal day when an agent writes the first draft
- How code review, testing and on-call changed
- Which skills matter more now, and which don't



Who to book

- Staff engineer who led an AI coding rollout
- Ideally someone who also interviews new grads

~/payments

```
$ agent "fix the flaky retry test"  
  > test races the real clock  
  > swapped in a fake clock +12 -4  
  
$ pytest -q  
214 passed  
  
✓ opened PR #4817  
  > waiting on review from @maya
```

An agent's first-draft fix, before review

Ask Them

What do you look for in new grads now?

Then: What would you learn if you were a sophomore today?

Talk Spotlight: Betting on Everything

AEZ

Kalshi and Polymarket traded \$53B in July, more than 10x last September

WEATHER · BERKELEY

Will it rain on Big Game day?

Yes

34¢

No

67¢

35¢

3,905

34¢

8,112

Every contract is a yes or no price between 1¢ and 99¢



What they'd cover

- Pricing a contract that has never traded before
- Making markets when most of the volume is sports
- Matching engines, settlement and the CFTC



Who to book

- Trader or quant on an event-contracts desk
- Susquehanna was Kalshi's first institutional market maker

Ask Them

Where does retail lose the most edge?

Then: Which markets do you refuse to make?

Talk Spotlight: The AWS DNS Bug

AEZ

One empty DNS record inside AWS took down Venmo, Snapchat, Fortnite and Canvas



What they'd cover

- How one empty DNS record cascaded for 14 hours
- What a 3am incident call sounds like
- Writing a postmortem that doesn't blame a person



Who to book

- SRE or incident commander at a cloud provider
- Anyone on call at a big consumer app that day

AWS Health · us-east-1

● DynamoDB error rates

- | | |
|----------|------------------------------|
| 11:48 PM | ● Errors start |
| 12:26 AM | ● Empty DNS record found |
| 2:40 AM | ● DNS fixed, backlog remains |
| 2:20 PM | ● Fully recovered |

Oct 19–20, 2025, from the first alarm to recovery (PDT)

Ask Them

What do you check first when everything is red?

Then: What changed in your runbooks afterward?

Talk Spotlight: The AI Scribe

AEZ

Epic shipped its own AI charting in February, while Abridge kept growing

Draft note

AI draft · review

Subjective

Chest tightness on stairs, mostly AM. Takes lisinopril 10 mg daily.

Assessment

Exertional chest discomfort. BP above goal.

Plan

ECG today. Order stress test.

Review & sign

What the doctor sees after a 15-minute visit



What they'd cover

- Turning a messy 15-minute visit into a clean note
- Keeping a doctor in the loop without slowing them down
- Why Epic built its own instead of buying



Who to book

- ML or product engineer at Abridge or a similar startup
- A clinical informatics lead at a health system like UCSF

Ask Them

What happens when it mishears a dosage?

Then: How do doctors decide whether to trust it?

05

Next Steps

● Finding Speakers

● The Plan

Finding Speakers

Warm intros first, then engineering blogs

AEZ



Alumni

- LinkedIn → UC Berkeley → Alumni, then filter by company and "Engineering" or "Finance"
- Ask older actives who they know. A warm intro gets a yes way faster than a cold DM



LinkedIn Search

"low latency" AND FPGA AND Berkeley

"site reliability" AND Berkeley

"ambient" AND engineer AND health



Engineering Blogs

- If someone wrote the engineering post, they already have the talk
- Discord, Cloudflare, Stripe, Jane Street, HRT and OpenAI all name their authors

Sample DM

"Hi Sam, I'm in AEZ at Berkeley and loved your post on moving your message store to ScyllaDB. Would you give a 30-minute talk to about 40 students who want to do this kind of work? Zoom or in person, any week this fall."

The Plan

What happens after this meeting

AEZ



Vote for your
top 3 talks

The form goes out after this meeting



Reach out,
alumni first

PD chairs DM alumni first



Book four talks
for the semester

About one talk every three weeks